LAB 7 – FUZZING FOR INPUT VALIDATION BUGS

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1. Execution of Original Application

```
C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>python processor.p
y
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): LAT
Base Price: $ 4.5

==== Running calculate_average_shot_cost ====
Enter CSV of shot prices (e.g. 1.0, 1.5, 2.0): 1.0, 1.5, 2.0
Average Shot Price: $ 1.5

==== Running parse_order_modifiers ====
Enter modifiers (e.g. milk:oat;syrup:vanilla): milk:oat;syrup:vanilla
Parsed Modifiers: {'milk': 'oat', 'syrup': 'vanilla'}

C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>
```

2. Fuzz Testing – Failed Test Cases

```
    | AssertionError: parse_order_modifiers crashed with: 'NoneType' object
has no attribute 'split'
    | assert False
    | Failing test case: test_parse_order_modifiers_safe(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7\test
_processor.py:26
    +------------------------------------
========================= short test summary info =========================
FAILED test_processor.py::test_lookup_drink_code_no_crash - ExceptionGroup:
Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_calculate_average_shot_cost_safe - ExceptionG
roup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_parse_order_modifiers_safe - ExceptionGroup:
Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================ 3 failed in 3.80s ============================

C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>

C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>
```

3. Execution of Fixed Application

```
C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>python processor.p
y
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): LAT
Base Price: $ 4.5

==== Running calculate_average_shot_cost ====
Enter CSV of shot prices (e.g. 1.0, 1.5, 2.0): 1.0, 1.5, 2.0
Average Shot Price: $ 1.5

==== Running parse_order_modifiers ====
Enter modifiers (e.g. milk:oat;syrup:vanilla): milk:oat;syrup:vanilla
Parsed Modifiers: {'milk': 'oat', 'syrup': 'vanilla'}

C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>
```

4. Fuzz Testing – Successful Test Cases

```
C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>python -m pytest t
est_processor.py
============================= test session starts =============================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.8, locust-2.46.6,
 cov-7.1.0
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 1.09s ==============================

C:\Users\palle\Downloads\L6_L7_03_coffee_shop_order\Lab 7>
```